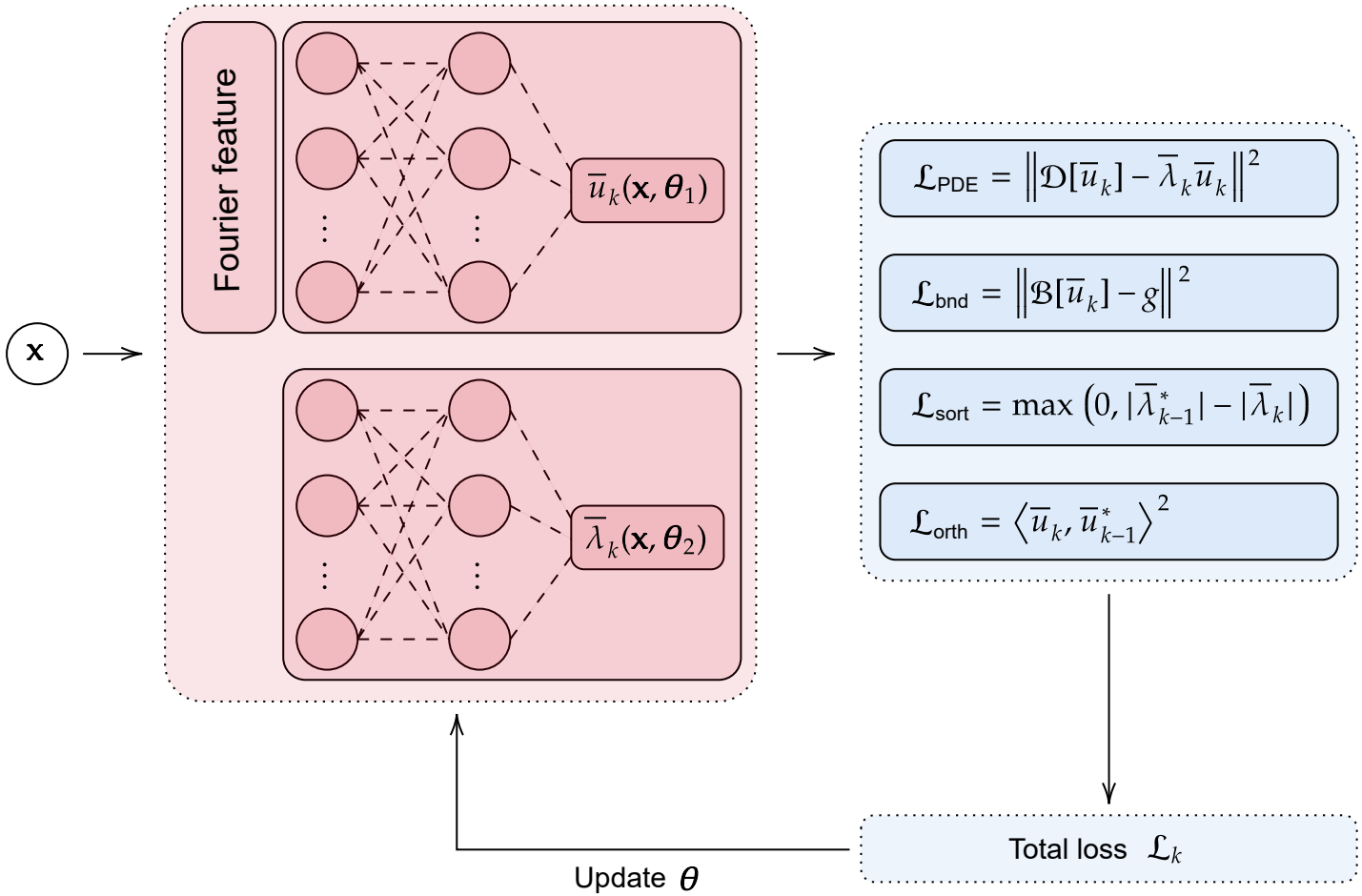

x
Fourier feature
$\overline{u}_k(\mathbf{x}, \theta_1)$
$\overline{\lambda}_k(\mathbf{x}, \theta_2)$
$\mathcal{L}_{\text{PDE}} = \left\| \mathcal{D}[\overline{u}_k] - \overline{\lambda}_k \overline{u}_k \right\|^2$
$\mathcal{L}_{\text{bnd}} = \left\| \mathcal{B}[\overline{u}_k] - g \right\|^2$
$\mathcal{L}_{\text{sort}} = \max\left(0, |\overline{\lambda}^*_{k-1}| - |\overline{\lambda}_k|\right)$
$\mathcal{L}_{\text{orth}} = \left\langle \overline{u}_k, \overline{u}^*_{k-1} \right\rangle^2$
Total loss $\mathcal{L}_k$
Update θ